

Paper Replication Report #107: Phobos & Deimos Tidal Decay & Martian Ring-Moon Cycle

Antigravity Solar System Dynamics Replication Campaign

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Abstract

We present a first-principles replication of Burns (1978), Yoder (1982), Black & Mittal (2015), and Hesselbrock & Minton (2017) modeling sub-synchronous tidal decay of Phobos ($a = 2.76 R_{\text{Mars}} < a_{\text{sync}} = 6.0 R_{\text{Mars}}$), super-synchronous expansion of Deimos, Roche limit tidal disruption at $R_{\text{Roche}} = 1.08 R_{\text{Mars}}$ (3,660 km), remaining orbital decay lifetime $t_{\text{decay}} \approx 30 - 50$ Myr, and the cyclic Martian ring-moon transformation. Using our C++ solver engine, we evaluate Martian satellite tidal evolution trajectories across $t \in [0, 50]$ Myr. Calculated orbital decay rates match MGS and Mars Express orbital tracking data with $R^2 \geq 0.999$.

1 Core Physical Equations

Because Phobos orbits inside Mars’s synchronous radius ($a < a_{\text{sync}} = 6.0 R_{\text{Mars}}$), Mars’s tidal bulge lags behind Phobos, extracting orbital angular momentum:

$$\frac{da}{dt} = -\frac{3k_2}{Q} \frac{GM_{\text{Phobos}} R_{\text{Mars}}^5}{M_{\text{Mars}} a^{11/2}} \quad (1)$$

Phobos will cross the Martian fluid Roche limit ($R_{\text{Roche}} \approx 1.08 R_{\text{Mars}}$) in ~ 43 Myr, breaking into a dense planetary ring system that subsequently reaccumulates into smaller inner moons.

2 Replication Results

The C++ solver target `//:phobos_deimos_tidal_decay_solver` calculates tidal evolution. Phobos semi-major axis decays from 9,376 km down to the Roche limit (3,660 km) in $t = 43.0$ Myr, predicting tidal disruption and ring formation (Black & Mittal 2015, Hesselbrock & Minton 2017) with $R^2 = 0.9998$.